MAOZUR

COLLATZ POINCARE FRONTIER A FRONTIER B FRONTIER C FRONTIER D MAZUR TAO MONK LAGARIAS

$$\boxed{\mathscr{M} = \mathscr{C} \cap \mathscr{R} \cap \mathscr{P} \cap \mathscr{A} \cap \mathscr{B} \cap \mathscr{C}_{\mathbf{F}} \cap \mathscr{D} \cap \mathscr{T} \cap \mathscr{Z} \cap \mathscr{N} \cap \mathscr{L}}$$

$$\mathscr{M} \overset{?}{\Longrightarrow} d_{\log}(B) = 0 \xrightarrow{\text{MAZUR}} B = \varnothing \iff \text{COLLATZ}$$

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0 COLLATZ

TARGET

$$T(n)=\begin{cases}\dfrac{n}{2}, & n\equiv 0 \pmod 2,\\[2mm] \dfrac{3n+1}{2}, & n\equiv 1 \pmod 2,\end{cases}\qquad \varepsilon_j(n)=T^j(n)\bmod 2,\qquad s_k(n)=\sum_{j<k}\varepsilon_j(n). \tag{0.1}$$

$$2^kT^k(n)=3^{s_k(n)}n+B_k(n),\qquad B_k(n)=\sum_{j=0}^{k-1}\varepsilon_j 2^j 3^{s_k-s_{j+1}}. \tag{0.2}$$

$$T^k(r+2^kq)=T^k(r)+3^{s_k(r)}q. \tag{0.3}$$

$$2^k>3^{s_k(n)}\quad\Longrightarrow\quad\Big[T^k(n)<n\Big]\Longleftrightarrow\Big[B_k(n)<\big(2^k-3^{s_k(n)}\big)n\Big]. \tag{0.4}$$

$$\Gamma=\{n\in\mathbb{N}_{>0}:\exists j\ge 0,\ T^j(n)=1\},\qquad B=\mathbb{N}_{>0}\setminus\Gamma. \tag{0.5}$$

$$m\sim_\Gamma n\Longleftrightarrow\exists i,j\ge 0:\ T^i(m)=T^j(n),\qquad \mathcal{E}=\{\min[n]_{\sim_\Gamma}:n\in B\}. \tag{0.6}$$

$$\boxed{\text{COLLATZ}\Longleftrightarrow B=\varnothing\Longleftrightarrow\mathcal{E}=\varnothing} \tag{0.7}$$

CROWN

REDUCTION

$$\mathrm{SYR}^a\big(\lambda 2^a3^b-1\big)=\frac{\lambda 3^{a+b}-1}{2^{\nu_2(\lambda 3^{a+b}-1)}}. \tag{0.8}$$

$$a+b=k\quad\Longrightarrow\quad\mathrm{SYR}^a\big(\lambda 2^a3^b-1\big)=Y_{\lambda,k},\qquad Y_{\lambda,k}:=\mathrm{ODD}(\lambda 3^k-1). \tag{0.9}$$

$$N_{\mathcal{E}}(X):=\#(\mathcal{E}\cap[1,X]),\qquad N_{\mathcal{E}}(X)\ll X^{\alpha},\qquad \alpha<1. \tag{0.10}$$

SUPERDYADIC

REDUCTION

$$X_j=2^{2^j},\qquad I_j=(X_{j-1},X_j],\qquad X_{j+1}=X_j^2. \tag{0.11}$$

1 MONK

INPUT / REDUCTION

$$\boxed{\textbf{COLLATZ} \iff \exists A, B_0 \in \mathbb{N}_{>0}\ \forall t \geq 0 :\ A + B_0 t \in \Gamma} \tag{1.1}$$

$$P(t) = 768t + 763. \tag{1.2}$$

$$P(9u+8) \sim_\Gamma P(8u+7), \qquad 8u+7 < 9u+8. \tag{1.3}$$

$$H^{27}\Big(P(2^{19}3^9 u + 4)\Big) = 3^{24}u + 137 = H^{39}\Big(P(2^{31}u)\Big). \tag{1.4}$$

$$P(2^{19}3^9 u + 4) \sim_\Gamma P(2^{31}u). \tag{1.5}$$

$$\mathsf{R}_{763}: \quad \forall t > 0 :\ \Big[t \not\equiv 8 \pmod 9\Big] \Longrightarrow \exists r < t :\ P(t) \sim_\Gamma P(r). \tag{1.6}$$

$$\boxed{\mathsf{R}_{763} \Longrightarrow \forall t \geq 0 :\ P(t) \sim_\Gamma 763 \Longrightarrow \textbf{COLLATZ}} \tag{1.7}$$

LAGARIAS

INPUT / FINITE CERTIFICATE

$$L_k(a) = \#\partial \mathcal{T}_k^*(a), \qquad M_K = \max_{3 \nmid a} L_K(a). \tag{1.8}$$

$$\exists K \geq 1 :\ M_K < 2^K. \tag{1.9}$$

$$2747 \leq L_{30}(a) \leq 9788 < 2^{30}, \qquad 3 \nmid a. \tag{1.10}$$

$$\Delta_L := \frac{1}{30}\log_2 \frac{2^{30}}{9788} > 0, \qquad L_k(a) \lesssim_{\exp} 2^{(1-\Delta_L)k}. \tag{1.11}$$

2 TAO

INPUT

$$\nu_X(A) := \frac{1}{\log X}\sum_{n\le X}\frac{1_A(n)}{n}. \tag{2.1}$$

$$S_Y := \left\{ n \in \mathbb{N}_{>0} : \min_{k\ge 0} H^k(n) > Y \right\}. \tag{2.2}$$

$$\boxed{\exists A_0,\gamma>0\ \forall X,Y\ge 2: \qquad \nu_X(S_Y) \le \frac{A_0}{(\log Y)^\gamma}} \tag{2.3}$$

$$g(n)=\lfloor \sqrt{n}\rfloor, \qquad \sigma(n)=\inf\{t\ge 0 : C^t(n)\le g(n)\}. \tag{2.4}$$

$$J(n)=\begin{cases} C^{\sigma(n)}(n), & \sigma(n)<\infty,\\ \dagger, & \sigma(n)=\infty, \end{cases} \qquad \mathcal{P}\delta_n=\delta_{J(n)}. \tag{2.5}$$

$$a_k(X) := (\mathcal{Q}^k\tilde{\mu}_X)(\dagger), \qquad \forall k\ge 0: \quad \lim_{X\to\infty} a_k(X)=0. \tag{2.6}$$

FRONTIER A

OPEN

$$\boxed{\liminf_{X\to\infty}\sup_{k\ge 0} a_k(X) \overset{?}{\le} \sup_{k\ge 0}\liminf_{X\to\infty} a_k(X)} \tag{2.7}$$

$$\textbf{TAO} + \textbf{FRONTIER A} \quad \overset{?}{\Longrightarrow} \quad \liminf_{X\to\infty}\tilde{\mu}_X(B)=0. \tag{2.8}$$

MAZUR

INPUT

$$d_{\log}(B) := \lim_{X\to\infty}\nu_X(B). \tag{2.9}$$

$$\boxed{d_{\log}(B)=0 \quad\Longrightarrow\quad B=\varnothing} \tag{2.10}$$

$$d_{\log}(B)=0 \quad \xRightarrow{\textbf{MAZUR}} \quad B=\varnothing \quad\Longleftrightarrow\quad \textbf{COLLATZ}. \tag{2.11}$$

3 POINCARE

ALGEBRA

$$\Delta F_X(t) = F_X(t) - F_X(t+1), \qquad \Delta^2 F_X(t) = F_X(t) - 2F_X(t+1) + F_X(t+2). \tag{3.1}$$

$$d_t(X) = \frac{K}{2}\,\Delta^2 F_X(t). \tag{3.2}$$

$$\Lambda_X := \frac{\|V_X\|_1}{\|(I-\mathcal{Q})V_X\|_1}, \qquad \|V_X\|_1 = \Lambda_X\,\|(I-\mathcal{Q})V_X\|_1. \tag{3.3}$$

FRONTIER B

OPEN

$$R_k = d_k + R_{k+1}, \qquad R_k, d_k \geq 0. \tag{3.4}$$

$$q_r := \frac{r}{r+1} < 1, \qquad \boxed{R_{k+1} \overset{?}{\leq} q_r R_k}. \tag{3.5}$$

$$\text{FRONTIER B} \Longrightarrow R_{k+m} \leq q_r^m R_k, \qquad d_k \geq (1-q_r)R_k = \frac{R_k}{r+1}. \tag{3.6}$$

FRONTIER C

OPEN

$$\boxed{\Lambda_X = \frac{\|V_X\|_1}{\|(I-\mathcal{Q})V_X\|_1} \overset{?}{\leq} K - s_e}. \tag{3.7}$$

$$\text{FRONTIER C} \iff \|V_X\|_1 \overset{?}{\leq} (K-s_e)\,\|(I-\mathcal{Q})V_X\|_1. \tag{3.8}$$

4 FRONTIER D

OPEN

$$\boxed{\exists k \geq 1: \qquad \liminf_{s \downarrow 0} \int_0^\infty u^{k-1}\, \mathcal{H}_u\, \mathfrak{G}_B(u, 1+s+k)\, \mathrm{d}u < \infty} \tag{4.1}$$

$$\mathcal{S}(B) := \sum_{m \in B} \frac{2^{D(m)}}{m}. \tag{4.2}$$

$$\textbf{FRONTIER D} \quad \stackrel{?}{\Longrightarrow} \quad \mathcal{S}(B) < \infty. \tag{4.3}$$

$$\mathcal{S}(B) < \infty \quad \stackrel{?}{\Longrightarrow} \quad B = \varnothing \quad \Longleftrightarrow \quad \textbf{COLLATZ.} \tag{4.4}$$

MERGE

ALGEBRA

$$\begin{aligned}\mathscr{M} := \mathscr{C}_{\textbf{COLLATZ}} \cap \mathscr{C}_{\textbf{CROWN}} \cap \mathscr{C}_{\textbf{SUPERDYADIC}} \cap \mathscr{C}_{\textbf{POINCARE}} \cap \mathscr{C}_{\textbf{TAO}} \cap \mathscr{C}_{\textbf{MONK}} \cap \mathscr{C}_{\textbf{LAGARIAS}} \\ \cap \mathscr{C}_{\textbf{FRONTIER A}} \cap \mathscr{C}_{\textbf{FRONTIER B}} \cap \mathscr{C}_{\textbf{FRONTIER C}} \cap \mathscr{C}_{\textbf{FRONTIER D}} \cap \mathscr{C}_{\textbf{MAZUR}}.\end{aligned} \tag{4.5}$$

$$\Omega_{\textbf{MAOZUR}} := \{\omega : \omega \models \mathscr{M}\}. \tag{4.6}$$

$$\begin{aligned}&\textbf{TAO} \wedge \textbf{MONK} \wedge \textbf{LAGARIAS} \wedge \textbf{CROWN} \wedge \textbf{SUPERDYADIC} \wedge \textbf{POINCARE} \\ &\quad \wedge \textbf{FRONTIER A} \wedge \textbf{FRONTIER B} \wedge \textbf{FRONTIER C} \wedge \textbf{FRONTIER D} \quad \stackrel{?}{\Longrightarrow} \quad d_{\log}(B) = 0 \\ &\qquad \stackrel{\textbf{MAZUR}}{\Longrightarrow} \quad B = \varnothing \quad \Longleftrightarrow \quad \textbf{COLLATZ.}\end{aligned} \tag{4.7}$$

$$\boxed{\begin{aligned}\Omega_{\textbf{SEARCH}} &= \Omega_{\textbf{MAOZUR}} \\ &= \bigcap_{\textbf{X}} \Omega_{\textbf{X}} \\ &= \bigcap \textbf{REDUCTIONS}\end{aligned}} \tag{4.8}$$